

StyCLIP: StyleTransfer in Few-Shot Learning

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Introduction & Motivation

- **Challenge:** Visual artist classification is crucial for art history and content analysis, especially with growing digital collections.
- **Focus:** Few-shot learning (FSL) for artist classification – identifying artists with limited labeled examples.
- **Problem:**
 - Artistic style involves subtle visual cues (brushstrokes, color palettes, composition).
 - CLIP-based FSL methods excel at semantics but might overlook fine-grained stylistic nuances.
- **Hypothesis:** Integrating explicit style representations (Gram matrices) into CLIP could enhance artist-specific pattern capture.
- **Proposal:** StyCLIP, a CLIP extension incorporating style features.

Related Work

Style Transfer

- Gatys et al. [3]: Deep networks encode style in intermediate layers; Gram matrices capture style.

Few-Shot Learning (FSL) & Meta-Learning

- Prototypical Networks [6]: Learn to learn from small samples.

Contrastive Language-Image Pretraining (CLIP)

- CLIP [5]: Shared embedding space for images and text.
- TIP-Adapter [8]: Non-parametric cache model for FSL.
- APE [9]: Adaptive prior refinement of CLIP features.
- GDA-CLIP [7]: Gradient-based adaptation.
- TIMO [4]: Mutual guidance between text/image modalities.

Dataset: ArtGraph Subset

- Derived from ArtGraph dataset [2] ($>100,000$ paintings).
- Two non-overlapping subsets created:

ArtGraph Few-Shot Subset (Evaluation)

- **Style Selection:** Top 10 styles by artwork count.
- **Artist Filtering:** 30–50 artworks/style (up to 10 artists/style).
- **Splits:** 80% support, 10% validation, 10% query.

ArtGraph StyCLIP Subset (Training)

- Complementary artists (16–50 artworks).
- Excluded from few-shot subset (no data leakage).

StyCLIP: Architecture Overview

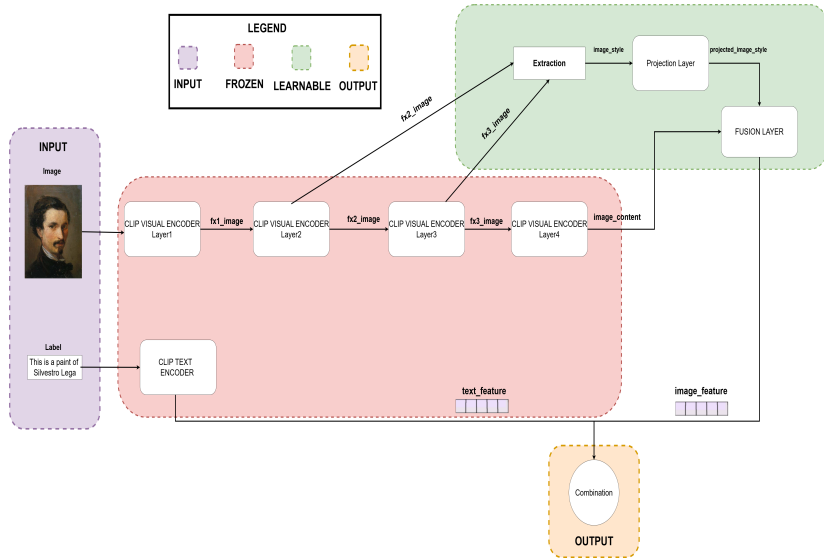


Figure: StyCLIP Architecture Overview

StyCLIP: Key Components

- **Core Idea:** Extend CLIP with explicit style-aware features via Gram matrices
- **Three Key Components:**
 - ① **Gram Matrix Extraction:** Capture style correlations from intermediate visual layers (ResNet-50 blocks)
 - ② **Style Feature Projection:** Reduce dimensionality of Gram matrix representations
 - ③ **Fusion Adapter:** Integrate semantic (CLIP) and style features through bottleneck architecture

StyCLIP: Training Strategy

- **Frozen Backbone:** Original CLIP (RN50) visual and text encoders are frozen.
- **Trainable Components:** Only the new style projection layers and fusion adapter are optimized.
- **Rationale:**
 - Preserve pre-trained semantic knowledge from CLIP.
 - Reduce computational cost.
 - Enable efficient fine-tuning on artistic datasets.

Hyperparameter Tuning

- Optuna [1] used for:
 - Fusion bottleneck dimension $\{64, 128, 256\}$
 - Dropout rate $[0.0, 0.5]$
 - Learning rate log-uniform $[10^{-5}, 10^{-3}]$
 - Weight decay log-uniform $[10^{-6}, 10^{-3}]$

Experimental Setup

Meta-Learning Training for StyCLIP

- Prototypical network approach.
- Episodic training: N-way (N=10 artists) classification.
- K support examples, Q=3 query examples per artist per episode.
- Two StyCLIP training configurations:
 - **StyCLIP FSL1**: Trained with K=1 support shot.
 - **StyCLIP FSL4**: Trained with K=4 support shots.

Evaluation Protocol

- N-way classification (all available artists in the test subset).
- Tested across $K \in \{1, 2, 4, 8, 16\}$ shots for support set.
- **Statistical Robustness**: 10 different random seeds.
- **Metrics**: Accuracy, Macro-Precision, Macro-Recall, Macro-F1.

Baseline Methods & Backbones

Few-Shot Adaptation Methods

- CLIP + TIP-Adapter [8]
- CLIP + APE [9]
- CLIP + GDA-CLIP [7]
- CLIP + TIMO [4]
- CLIP + TIMO-S

Backbones Compared

- **CLIP RN50**: Standard CLIP ResNet-50.
- **StyCLIP FSL1**: Our custom backbone, trained with 1-shot.
- **StyCLIP FSL4**: Our custom backbone, trained with 4-shots.

An example episode for RN50:

Hyperparameter Optimization Highlights

StyCLIP FSL1 (1-shot trained)

- Bottleneck: 64
- LR: 6.01×10^{-5}
- Dropout: 0.0
- Weight Decay: 4.83×10^{-4}
- Valid Acc: 57.07%

StyCLIP FSL4 (4-shot trained)

- Bottleneck: 128
- LR: 1.02×10^{-5}
- Dropout: 0.1
- Weight Decay: 1.03×10^{-6}
- Valid Acc: 60.53%

Example Episode for RN50

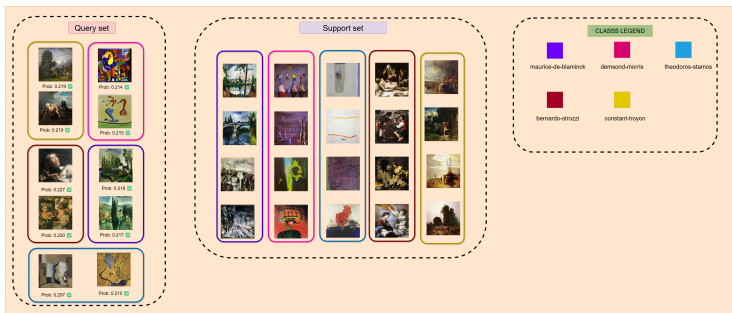


Figure: Example episode: 5-Way, 4-Shots ,10 query examples

Key Evaluation Results

Main Finding: Integrating explicit style-aware features (StyCLIP) *consistently underperforms* standard CLIP variants.

- Standard CLIP with RN50 backbone is the best performing backbone.
- StyCLIP backbones (StyCLIP_FSL1, StyCLIP_FSL4) significantly degrade performance for all FSL methods.
- TIMO and TIMO-S (our adaptation logic) are the best FSL methods when using the standard RN50 backbone.

Example: 16-Shot Classification Accuracy (%)

Model	Backbone	ACC (%)
TIMO	StyCLIP_FSL1	37.00
TIMO	StyCLIP_FSL4	43.27
TIMO	RN50	73.36
TIMO_S	StyCLIP_FSL1	39.20
TIMO_S	StyCLIP_FSL4	48.75
TIMO_S	RN50	73.46

Statistical Analysis Summary

- Paired t-tests confirmed the significance of performance differences.
- **Key Observation:** Standard CLIP (RN50) significantly outperforms StyCLIP-trained backbones across nearly all configurations and few-shot methods.
- Example: For 16-shot with TIMO_S:
 - RN50 (73.46%) vs StyCLIP_FSL4 (48.75%) $\rightarrow \approx 25$ percentage point difference, statistically significant.

$\Rightarrow$ The architectural changes in StyCLIP for explicit style modeling did not yield benefits and often hindered performance.

Conclusion & Discussion

- **Main Finding:** Integrating explicit style-aware features via Gram matrices (StyCLIP) did **not improve** few-shot artist classification performance.
- **Performance:** Standard CLIP (RN50 backbone) consistently outperformed StyCLIP variants.
- **Implication:** Explicit Gram matrix-based style modeling might interfere with CLIP's pre-trained semantic representations rather than enhance them for this task.
- **Future Directions:** Alternative approaches like attention-based style integration or domain-specific fine-tuning (without architectural changes) might be more fruitful.

Acknowledgments

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The code for this project is available on GitHub:

<https://github.com/Fonty02/ComputerVision/tree/main/TIMO>

Thank You!

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